

```

clc;

clear;

close all;

% EX NO: 1

% Butterworth IIR Low-Pass Filter Design and Analysis

% Date:

% Specifications

Fs = 10000;    % Sampling Frequency (Hz)

Fp = 1000;     % Passband Frequency (Hz)

Fst = 1500;    % Stopband Frequency (Hz)

Rp = 1;        % Passband Ripple (dB)

Rs = 60;       % Stopband Attenuation (dB)

% Calculate Minimum Filter Order

[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);

% Design Butterworth Low-Pass Filter

[b, a] = butter(N, Wn, 'low');

% Magnitude and Phase Response

figure;

freqz(b, a, 1024, Fs);

title('Magnitude and Phase Response');

% Impulse Response

figure;

impz(b, a, 50, Fs);

```

```
title('Impulse Response');
```

```
% Step Response
```

```
figure;
```

```
stepz(b, a);
```

```
title('Step Response');
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b, a);
```

```
title('Pole-Zero Plot');
```

```
% Display Results
```

```
fprintf('Minimum Filter Order (N) = %d\n', N);
```

```
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);
```

```
fprintf('Actual Cutoff Frequency = %.2f Hz\n', Wn * (Fs/2));
```



